

DP-HHH Fixed Proof

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1 Fixed Privacy Proof

Shown below is the updated algorithm. Instead of a single $\gamma \leftarrow \text{Laplace}(\frac{\varepsilon}{4})$ we use a new v_p per node. This does not meaningfully change the accuracy of the algorithm as the noise added per node is still independent of height of the hierarchy.

Algorithm 1: Non-Streaming DP-HHH Detection

Input: Data X of size n over hierarchy $\mathcal{H}$ with height h , Privacy parameter $\varepsilon \in (0, \log n)$, $\delta = \text{negl}(n)$, Threshold $\tau > 0$

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1   $C := O(\log \frac{h}{\delta})$ 
2  If  $\tau < C + 1$ , output  $(\{\}, 0)$  and end program.
3   $\mathcal{S} = \{\}$ 
4  for  $i = h, \dots, 1$  do
5       $\mathcal{A}_i = \{p \in \mathcal{H} \mid \text{Level}(p) = i\}$ 
6      for  $p \in \mathcal{A}_i$  do
7          if  $F_{\mathcal{S}}(p) = 0$  then continue.
8           $w_p \leftarrow \text{Laplace}(\frac{\varepsilon}{4})$ 
9           $v_p \leftarrow \text{Laplace}(\frac{\varepsilon}{4})$ 
10         if  $F_{\mathcal{S}}(p) + w_p + v_p \geq \tau$  then
11              $\mathcal{S} = \mathcal{S} \cup \{p\}$ 
12              $\tilde{F}_{\mathcal{S}}(p) = F_{\mathcal{S}}(p) + \text{Laplace}(4/\varepsilon)$ 
13         end if
14     end for
15 end for
16 Output  $\mathcal{S}$  and  $\{\tilde{F}_{\mathcal{S}}(p)\}_{p \in \mathcal{S}}$ .
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Algorithm 1: Non-Streaming DP-HHH Detection

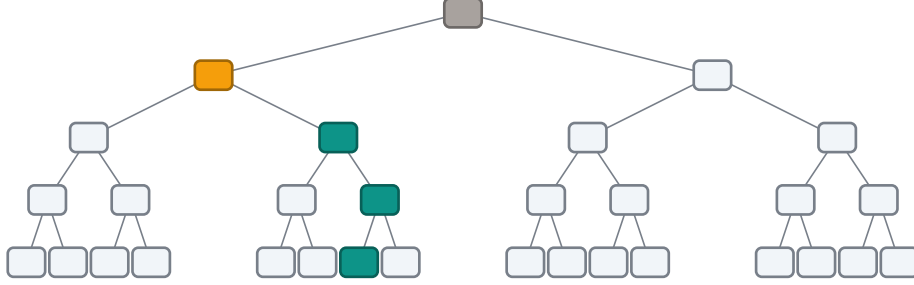


Figure 1: The figure above describes the partitioning of $\mathcal{H}$ assuming the neighbouring element x' is *not* isolated. The unfilled nodes ($\mathcal{J}_{\text{Unrelated}}$) denote all the nodes whose residual frequencies are unaffected by the presence of x' . The nodes in teal are the prefixes whose conditional frequencies include x' , but their noisy conditional frequency (the noisy-threshold step of [Algorithm 1](#)) is below τ . The node in amber is referred to as the inflection node. It is the first prefix of x' whose noisy count crosses the threshold. Together, the teal and amber nodes form $\mathcal{J}_{\text{Active}}$. For all nodes that are generalisations of the inflection node (shown in grey), their conditional count excludes any influence of x' , from the definition of conditional count ($\mathcal{J}_{\text{After}}$).

Definition 1.1 (Partition induced by the output) Fix neighbouring data sets X and $X' = X \cup \{x'\}$. Let $\vec{a} \in \{\perp, \top\}^{|\mathcal{H}|}$ denote any arbitrary output of [Algorithm 1](#). Let $p^* := p^*(\vec{a})$ be the lowest parent of x' such that $a_{p^*} = \top$ i.e.

$$p^* := p^*(\vec{a}) = \arg \min_{\substack{p \in \mathcal{H} \text{ s. t.} \\ x' \succeq p \\ a_p = \top}} \text{Level}(p)$$

Then, we can partition $\mathcal{H}$ with the following sets based on p^* .

$$\begin{aligned} \mathcal{J}_{\text{Unrelated}} &:= \{p \in \mathcal{H} : x' \not\succeq p\} \\ \mathcal{J}_{\text{After}} &:= \{p \in \mathcal{H} : p^* \succ v\} \\ \mathcal{J}_{\text{Active}} &:= \mathcal{H} \setminus (\mathcal{J}_{\text{Unrelated}} \cup \mathcal{J}_{\text{After}}) \end{aligned}$$

Theorem 1.2 (Privacy) [Algorithm 1](#) is (ξ, δ) -DP.

Proof. Fix datasets X and $X' := X \cup \{x'\}$. Fix output $\vec{a} := (a_p)_{p \in \mathcal{H}}$ where $a_p \in \{\perp, \top\}$. Define $M := |\mathcal{H}|$. Let $(\beta_1, \dots, \beta_M) \leftarrow \mathbb{A}(X)$ and $(\beta'_1, \dots, \beta'_M) \leftarrow \mathbb{A}(X')$ be the output of [Algorithm 1](#) on inputs X and X' respectively with randomness over the samples $(\vec{w}, \vec{v}) \leftarrow \mathbb{Laplace}(\frac{\epsilon}{4})^{|\mathcal{H}|}$.

For any given $p \in \mathcal{H}$, let $\text{Succ}(p) := \{q \in \mathcal{H} : q \succ p\}$ denote the strict successors of p in $\mathcal{H}$ and let $\text{Par}(q) := \{p \in \mathcal{H} : q \succeq p\}$ denote the ancestors of q in $\mathcal{H}$ (including q).

Now for a fixed $\vec{a}$, and let $p^* := p^*(\vec{a})$ as defined in [Definition 1.1](#). Based on p^* , we partition $\mathcal{H}$ into three sets — $\mathcal{J}_{\text{Unrelated}}$, $\mathcal{J}_{\text{Active}}$, $\mathcal{J}_{\text{After}}$ as described by [Figure 1](#).

Let $\mathcal{R} := \mathcal{H} \setminus \mathcal{J}_{\text{Unrelated}} = \mathcal{J}_{\text{Active}} \cup \mathcal{J}_{\text{After}}$. We first split off the $\mathcal{J}_{\text{Unrelated}}$ coordinates. By Bayes rule, we have

$$\Pr[\vec{\beta} = \vec{a}] = \Pr[\vec{\beta}_{\mathcal{J}_{\text{Unrelated}}} = \vec{a}_{\mathcal{J}_{\text{Unrelated}}}] \cdot \Pr[\vec{\beta}_{\mathcal{R}} = \vec{a}_{\mathcal{R}} \mid \vec{\beta}_{\mathcal{J}_{\text{Unrelated}}} = \vec{a}_{\mathcal{J}_{\text{Unrelated}}}] \quad (1)$$

Focusing on the left hand term of the product in [Equation 1](#), we focus only on $p \in \mathcal{J}_{\text{Unrelated}}$. As the algorithm proceeds bottom up, we have for any $p \in \mathcal{J}_{\text{Unrelated}}$,

$$\begin{aligned} \Pr[\beta_p = a_p \wedge \vec{\beta}_{\text{Succ}(p)} = \vec{a}_{\text{Succ}(p)}] \\ = \Pr[\beta_p = a_p \mid \vec{\beta}_{\text{Succ}(p)} = \vec{a}_{\text{Succ}(p)}] \cdot \Pr[\vec{\beta}_{\text{Succ}(p)} = \vec{a}_{\text{Succ}(p)}]. \end{aligned} \quad (2)$$

The corresponding local factor is the same under X' :

$$\Pr[\beta_p = a_p \mid \vec{\beta}_{\text{Succ}(p)} = \vec{a}_{\text{Succ}(p)}] = \Pr[\beta'_p = a_p \mid \vec{\beta}'_{\text{Succ}(p)} = \vec{a}_{\text{Succ}(p)}]. \quad (3)$$

To justify [Equation 3](#), condition on the event $\vec{\beta}_{\text{Succ}(p)} = \vec{a}_{\text{Succ}(p)}$. This fixes exactly which already-processed successors of p have been selected into $\mathcal{S}$, namely those $q \in \text{Succ}(p)$ with $a_q = \top$. Therefore the threshold test at p is determined by the same fresh noise (w_p, v_p) and by the residual count of p after removing this fixed set of selected successors. Since $p \in \mathcal{J}_{\text{Unrelated}}$, no element on which X and X' differ lies below p ; hence this residual count is identical under X and X' (as the unconditional counts are equal).

For brevity of notation let the event E denote the event that $\vec{\beta}'_{\mathcal{J}_{\text{Unrelated}}} = \vec{a}_{\mathcal{J}_{\text{Unrelated}}}$. Applying [Equation 3](#) bottom up through the hierarchy over $\mathcal{J}_{\text{Unrelated}}$ gives

$$\Pr[E] := \Pr[\vec{\beta}_{\mathcal{J}_{\text{Unrelated}}} = \vec{a}_{\mathcal{J}_{\text{Unrelated}}}] = \Pr[\vec{\beta}'_{\mathcal{J}_{\text{Unrelated}}} = \vec{a}_{\mathcal{J}_{\text{Unrelated}}}] \quad (4)$$

Thus, it suffices to show to prove the theorem it suffices to show that

$$e^{-\epsilon} (\Pr[\vec{\beta}'_{\mathcal{R}} = \vec{a}_{\mathcal{R}} \mid E] - \delta) \leq \Pr[\vec{\beta}_{\mathcal{R}} = \vec{a}_{\mathcal{R}} \mid E] \leq e^{\epsilon} \Pr[\vec{\beta}'_{\mathcal{R}} = \vec{a}_{\mathcal{R}} \mid E] + \delta \quad (5)$$

Now by bayes rule and the structure of $\mathcal{H}$, we have

$$\Pr[\vec{\beta}_{\mathcal{R}} = \vec{a}_{\mathcal{R}} \mid E] = \Pr[\vec{\beta}_{\mathcal{J}_{\text{After}}} = \vec{a}_{\mathcal{J}_{\text{After}}} \mid E \wedge \vec{\beta}_{\mathcal{J}_{\text{Active}}} = \vec{a}_{\mathcal{J}_{\text{Active}}}] \Pr[\vec{\beta}_{\mathcal{J}_{\text{Active}}} = \vec{a}_{\mathcal{J}_{\text{Active}}}] \quad (6)$$

Notice by a similar argument above, $\vec{a}$ defines $\mathcal{S}$, and given E and p^* , we have

$$\Pr[\vec{\beta}_{\mathcal{J}_{\text{After}}} = \vec{a}_{\mathcal{J}_{\text{After}}} \mid E \wedge \vec{\beta}_{\mathcal{J}_{\text{Active}}} = \vec{a}_{\mathcal{J}_{\text{Active}}}] = \Pr[\vec{\beta}'_{\mathcal{J}_{\text{After}}} = \vec{a}_{\mathcal{J}_{\text{After}}} \mid E \wedge \vec{\beta}'_{\mathcal{J}_{\text{Active}}} = \vec{a}_{\mathcal{J}_{\text{Active}}} \mid E] \quad (7)$$

Thus to prove [Equation 5](#) it suffices to show

$$e^{-\epsilon} (\Pr[\vec{\beta}'_{\mathcal{J}_{\text{Active}}} = \vec{a}_{\mathcal{J}_{\text{Active}}} \mid E] - \delta) \leq \Pr[\vec{\beta}_{\mathcal{J}_{\text{Active}}} = \vec{a}_{\mathcal{J}_{\text{Active}}} \mid E] \quad (8)$$

$$< = e^{\epsilon} \Pr[\vec{\beta}'_{\mathcal{J}_{\text{Active}}} = \vec{a}_{\mathcal{J}_{\text{Active}}} \mid E] + \delta \quad (9)$$

Assume that $p^\star$ is not isolated. This happens with probability at least $1 - \delta$ (based on stability histograms argument of paper - that remains unchanged.)

Let $i^\star := \text{Level}(p^\star)$. We have by definition that $\vec{a}_{\mathcal{J}_{\text{Active}}} = (\perp)^{i^\star-1} \top$.

$$\begin{aligned} \Pr[\vec{\beta}_{\mathcal{J}_{\text{Active}}} = \vec{a}_{\mathcal{J}_{\text{Active}}} | E] &:= \Pr[\vec{\beta}_{p^\star} = a_{p^\star} | E] \prod_{p \in \mathcal{J}_{\text{Active}} \setminus \{p^\star\}} \Pr[\vec{\beta}_p = a_p | E] \\ &= \Pr[f_X(p^\star) + w_{p^\star} + v_{p^\star} \geq \tau | E] \prod_{p \in \mathcal{J}_{\text{Active}} \setminus \{p^\star\}} \Pr[f_X(p) + w_p + v_p < \tau | E] \\ &= \Pr[f_X(p^\star) + w_{p^\star} + v_{p^\star} \geq \tau | E] \Pr\left[\min_{p \in \mathcal{J}_{\text{Active}} \setminus \{p^\star\}} f_X(p) + w_p + v_p < \tau | E\right] \end{aligned}$$

Focus on the left term of the product

$$\Pr[f_X(p^\star) + w_{p^\star} + v_{p^\star} \geq \tau | E] = \Pr[f_X(p^\star) + w_{p^\star} + v_{p^\star} \geq \tau \mid w_{p^\star} \wedge E] \Pr[w_{p^\star}] \quad (10)$$

$$\leq e^{\frac{\xi}{4}} \Pr[f_X(p^\star) + w_{p^\star} + v_{p^\star} \geq \tau \mid w_{p^\star} \wedge E] \Pr[w_{p^\star}] \quad (11)$$

$$\leq e^{\frac{\xi}{4}} \Pr[f_{X'}(q) + w_q + v_q \geq \tau \mid E] \quad (12)$$

Now let $q \in \mathcal{J}_{\text{Active}} \setminus \{p^\star\}$ such that $f_q(X) + w_q + v_q \geq f_{p(X)} + w_p + v_p$ for all $p \in \mathcal{J}_{\text{Active}} \setminus \{p^\star\}$. Then

$$\Pr[f_X(q) + w_q + v_q < \tau \mid E] = \Pr[f_X(q) + w_q + v_q \geq \tau \mid v_q \wedge E] \Pr[v_q] \quad (13)$$

$$\leq e^{\frac{\xi}{4}} \Pr[f_X(p^\star) + w_q + v_q \geq \tau \mid v_q \wedge E] \Pr[v_q] \quad (14)$$

$$\leq e^{\frac{\xi}{4}} \Pr[f_{X'}(q) + w_q + v_q < \tau \mid E] \quad (15)$$

This completes the proof. $\square$